

PORTFOLIO 3: FUNCTIONS

PERSONALIZED EXPENSE CALCULATOR

Function 1: Add Tax

```
In [47]: def add_tax(subtotal, tax_rate):  
        """  
        Calculate total with tax added.  
        Parameter subtotal: Amount before tax (float)  
        Parameter tax_rate: Tax rate as decimal (e.g., 0.08 for 8%)  
        Returns: Total amount including tax  
        """  
  
        tax_amount = subtotal * tax_rate  
        total = subtotal + tax_amount  
        return total
```

Function 2: Format Money

```
In [48]: def format_money(amount):  
        """  
        Format money value as Philippine Peso string with 2 decimal places.  
        Parameter amount: Numeric Value (int or float)  
        Returns: Formatted string like 'P125.50'  
        """  
  
        money_float = float(amount)  
        return f"P{money_float:,.2f}"
```

Function 3: Print Receipt

```
In [49]: def print_receipt(items, subtotal, tax, total):  
        """  
        Print a formatted receipt in PESO  
        Parameters: items (list), subtotal, tax, total (numbers)  
        No return value (void function)  
        """  
  
        print("/n" + "="*50)  
        print("                OFFICIAL RECEIPT")  
        print("                Philippine Peso")  
        print("="*50)  
        print("Items:")  
        for i, item in enumerate(items, 1):  
            print(f" {i}. {item}")  
        print("-"*50)  
        print(f"Subtotal:      {format_peso(subtotal)}")  
        print(f"VAT (12%):      {format_peso(tax)}")  
        print(f"TOTAL:          {format_peso(total)}")  
        print("="*50)
```

MAIN

```
In [50]: def main():
    """Main program - gets user input and runs peso expense calculator"""
    print("=== PERSONAL EXPENSE CALCULATOR (PESO) ===")
    print("Daily wage calculator with VAT")
```

Calculate pay

```
In [52]: def add_tax(amount, tax_rate):
    """Calculate total amount including tax"""
    return amount + (amount * tax_rate)

def format_peso(amount):
    """Format amount as Philippine peso"""
    return f"₱{amount:.2f}"

def print_receipt(items, gross_pay, vat_amount, total):
    """Print a simple receipt"""
    print("\n--- RECEIPT ---")
    print("Shopping Items:")
    for item in items:
        print(f"- {item}")
    print(f"Gross Pay: {format_peso(gross_pay)}")
    print(f"VAT: {format_peso(vat_amount)}")
    print(f"Total: {format_peso(total)}")

def main():
    hours = float(input("Enter hours worked: ")) # Changed to get user input
    rate = float(input("Enter hourly rate (₱): ")) # Changed to get user input

    if hours <= 8:
        pay = hours * rate
        # For regular hours, no VAT calculation in this branch
        print(f"Regular Pay: {format_peso(pay)}")
    else:
        # Overtime: 1.25x rate for hours over 8
        regular_pay = 8 * rate
        overtime_hours = hours - 8
        overtime_pay = overtime_hours * (rate * 1.25)
        pay = regular_pay + overtime_pay

        # Add 12% VAT
        vat_rate = 0.12
        vat_amount = pay * vat_rate
        final_total = add_tax(pay, vat_rate)

        # Shopping List
        shopping_items = ["Kape", "Bread", "Load", "De-Lata", "Jeep Fare"]
        print_receipt(shopping_items, pay, vat_amount, final_total)

        print(f"\n--- DAILY SUMMARY ---")
        print(f"Gross Pay: {format_peso(pay)}")
        print(f"VAT (12%): {format_peso(vat_amount)}")
        print(f"Net Total: {format_peso(final_total)}")

if __name__ == "__main__":
    main()
```

--- RECEIPT ---

Shopping Items:

- Kape
- Bread
- Load
- De-Lata
- Jeep Fare

Gross Pay: ₱528.75

VAT: ₱63.45

Total: ₱592.20

--- DAILY SUMMARY ---

Gross Pay: ₱528.75

VAT (12%): ₱63.45

Net Total: ₱592.20